

VENKATRAMAN R

Data Scientist · Machine Learning Engineer · AI Engineer

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SUMMARY

Entry-level **Data Scientist** fluent across the full machine-learning lifecycle — statistics, algorithms, applied mathematics (linear algebra, calculus), modelling, explainability, data analytics, data visualization, and deployment — demonstrated through 7 production-grade projects shipped end to end. Strong in Python, SQL, generative AI (LangChain, FAISS, RAG), and classical ML (XGBoost, Random Forest); deployed models reach AUC-ROC 0.97 with audit-ready SHAP explainability. Analytical researcher with strong communication skills, problem-solving instincts, and tech savvy across modern programming and mathematical software stacks, who offsets a non-traditional academic path with a deep, shipped portfolio. Seeking Data Scientist, ML Engineering, or AI Engineering roles.

TECHNICAL SKILLS

Programming	Python, SQL
Math & Stats	Statistics, Linear Algebra, Calculus, Applied Mathematics, Probability, Algorithms, Hypothesis Testing, Analytical Methods, Mathematical Software, Business Rules
ML / AI	XGBoost, Random Forest, SMOTE, SHAP, Optuna, Scikit-Learn, TensorFlow/Keras, LSTM, Cross-Validation
GenAI / LLMs	LangChain (LCEL), FAISS, RAG, OpenAI API, Groq Llama-3.3-70B, HuggingFace, Prompt Engineering
Data Analytics & Viz	Pandas, NumPy, Power BI (DAX), Plotly, Matplotlib, Seaborn, Data Visualization, Streamlit, SQLite, Parquet, Power Query
Tools	Git, GitHub, Jupyter, VS Code, Google Colab, Advanced Excel

PROFESSIONAL EXPERIENCE

Data Science Capstone Engineer

Jan 2026 – Jun 2026

Boston Institute of Analytics — Professional Certification Program

- Researched and built **FraudGuard** (institute flagship capstone): trained an XGBoost classifier with Optuna (100 trials) and SMOTE balancing on imbalanced mobile-transaction data via rigorous analytical workflow, reaching AUC-ROC 0.97, F1 0.92, precision 0.93, recall 0.91; encoded fraud-cost asymmetry as business rules via F2-score threshold optimisation and added SHAP per-transaction explanations for audit compliance
- Designed **BankShield** capstone: an end-to-end Random Forest pipeline (EDA → feature engineering → SMOTE → threshold tuning → deployment) reaching AUC-ROC 0.97, precision 0.94, recall 0.89; surfaced transaction velocity as more predictive than amount; deployed a Streamlit app with single + batch CSV scoring and communicated findings through walkthroughs to non-technical reviewers

PROJECTS

AskMyData — GenAI Dataset Q&A Application

Live Demo · GitHub

Python · LangChain (LCEL) · FAISS · Groq Llama-3.3-70B · OpenAI · Streamlit

- Built a LangChain LCEL agent that auto-selects pandas or SQL to answer plain-English questions across 4 source formats (CSV, Excel, JSON, SQLite), with FAISS retrieving per-column schema context for accuracy
- Engineered a secure execution sandbox (whitelisted builtins, auto-retry), multi-turn memory, auto-generated Plotly visualizations per query, and a live Streamlit Cloud deployment

SalesPulse — Power BI Sales Intelligence Dashboard

GitHub

Power BI · DAX · Power Query · Star Schema · Data Visualization

- Authored 44 custom DAX measures (YoY, MTD/QTD/YTD, customer LTV, profit margin) over a star schema on \$12.63M of Global Superstore data, with executive, regional, product, and customer-segment drill-throughs

SalesPulse Forecaster — LSTM 12-Month Sales Forecasting

GitHub

TensorFlow · Keras · LSTM · SQLite · Streamlit · Statistics

- Built a 2-layer LSTM (128 → 64) generating recursive 12-month forecasts with confidence bands; backed by a SQLite pipeline and a Streamlit upload-to-export interface

DataHunt — Automated ETL Pipeline

GitHub

Python · Pandas · Parquet · openpyxl · Algorithms · Logging

- Designed a modular pipeline (Extractor → Cleaner → Storage → Reporter) outputting Parquet and Excel simultaneously, with fully configurable strategies and independently swappable modules

EDUCATION

Professional Certification — Data Science with Artificial Intelligence

Expected Jun 2026

Boston Institute of Analytics

Coursework: Statistics, Linear Algebra, Calculus, Probability, Algorithms, Machine Learning, Deep Learning, NLP, Generative AI & RAG, Power BI, SQL, EDA, Data Visualization. Capstones: FraudGuard (institute flagship) & BankShield.

Bachelor of Engineering (B.E.) — Automobile Engineering

2021 – 2024

Vels Institute of Science, Technology & Advanced Studies (VISTAS) · CGPA 7.25